

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

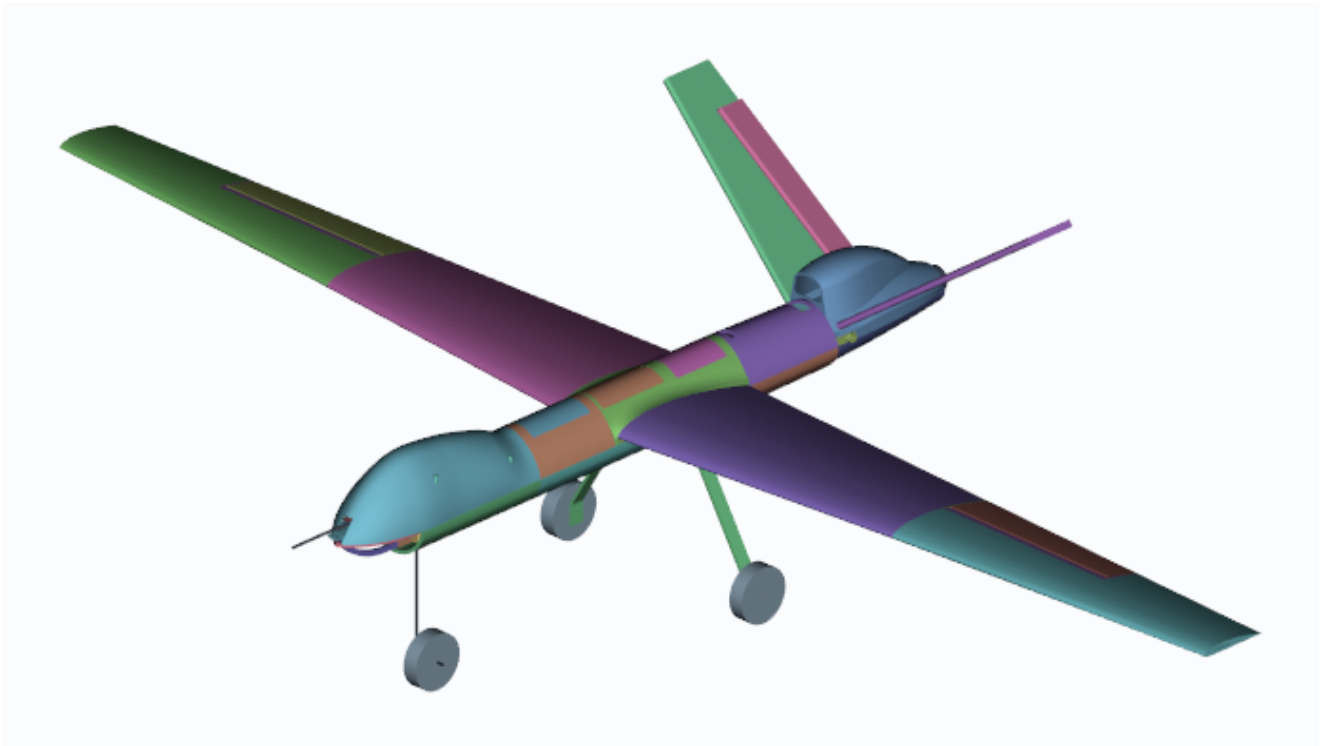
N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

01 / MQ9 airframe

Manufacturing Team / 2-metre RC model / Revision 2



Build the frame, body, mounts and controls with this guide. The Electronics Team guide covers wiring, sensor setup and calibration.

Build order	Finish each check before the next stage
1 / Coupons	Measure stock and test the cutting, printing and foam-forming processes.
2 / Skeleton	Dry-fit both wings, the fuselage and landing gear.
3 / Installation	Fit mounts, actual equipment, straps, controls and service openings.
4 / Covering	Form, fit and bond skins while retaining the specified access.
5 / Transfer	Record mass, fit and structural checks; hand over with the propeller removed.

Read the Weight and flight assessment chapter before the full batch. Digital geometry checks do not clear hand launch or flight; material, balance and propulsion checks remain.

02 / Shop files and stock

Folder	Workshop use
01_Plywood	CUT profiles by thickness, sheet layouts, quantities and fit coupons.
02_Foam	Developed patterns and forming references.
03_Print	Aircraft STLs, dimensions/quantities and printer coupon.
04_Hardware	Carbon schedule, metal gear geometry and forming references.
05_Assembly	Positioned STEP and assembly illustrations.
Print_Templates	Actual-size A4 cutting templates. Print at 100%, not Fit.

Machine work: laser/CNC plywood, print selected STLs, form and drill the metal gear. **Bench work:** knife-cut foam, form, bevel, glue, align, fasten and install linkages. The package does not remove these assembly steps.

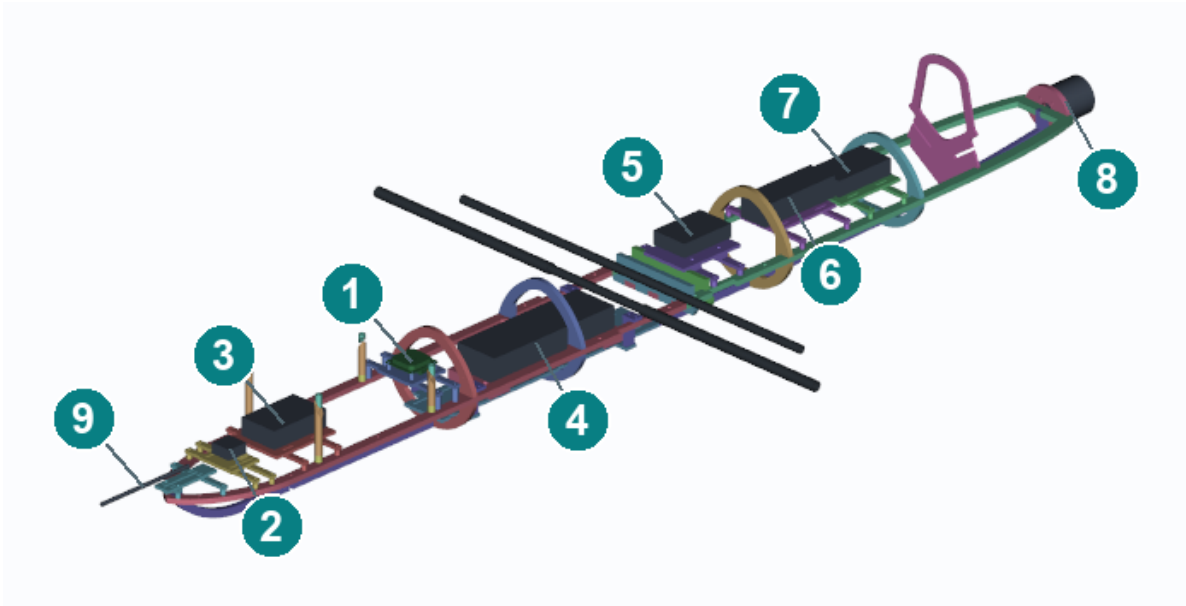
Use 5 mm FliteBoard Pro paper-faced XPS, 1000 x600 mm sheets. Use a knife or suitable CNC knife; laser processing requires the workshop to approve this exact material and extraction process.

Plywood: 24 pieces at nominal 5 mm, 6 at 3 mm and 7 at 2.5 mm. The two A44 ribs use the 4.990 mm geometry. Balsa is optional; it is not a substitute for the specified structural plywood.

The shop may re-nest 1220 x2440 mm layouts. Preserve scale, grain along long structural members and quantities. Do not scale a part to make a slot fit.

Tools: caliper, square, flat jig, knife, sanding blocks, clamps, drills/reamers, small nut drivers, long-nose pliers and a 1.5 mm ball-end hex driver. Make process coupons before cutting the aircraft set.

03 / Fuselage skeleton

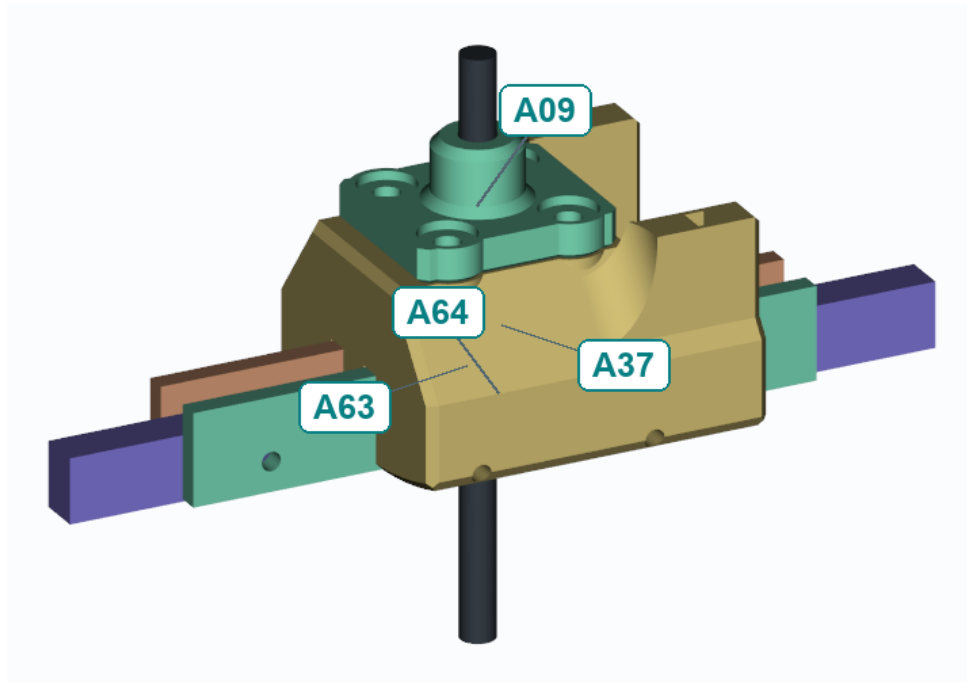


Assembly coordinates: X across the wings, Y toward the tail, Z up. Nose datum is Y=-439.65 mm. The STEP and mounting holes locate parts; this illustration shows equipment with covers removed.

1. Jig A62F/R on the centreline. Build their bolted nose bypass before bonding the frame.
2. Fit A13 and A43. Install A34 at Y=-130, A54 at Y=-5, A16 at Y=233.65 and A04 at Y=384.65. Seat the tabs fully without forcing.
3. Fit the main-gear box A25/A48 and bottom plate A47. Bond G03 below A47 with both M3 holes aligned. Fit tail bulkhead A18 and motor mount A26.
4. Dry-fit A59, A06, A12, both wings and all installation parts. Check equal left/right extents, square formers and straight centreline.
5. Bond after the complete dry-fit. Add modest fillets at gear, wing-root and motor joints. There is no separate A40 piece.

Keep the battery route open: the pack slides from the nose into its low central tray. Do not add a bulkhead, crossbar or wire bundle across this route.

04 / Nose gear and bypass

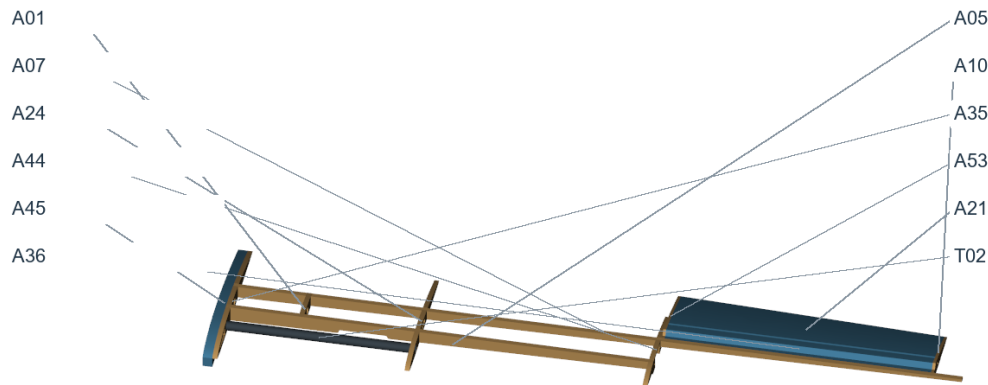


Below this bay, A63/A64 bridge the split spine around the steering shaft. Each cheek is 2.5 mm plywood, 72 x10.7 mm, with grain along its length.

1. Set A62F/R collinear with the **4.4 mm gap** centred at Y=-293.65. Fit the cheeks into A37, one per side of the 5 mm spine.
2. Install four transverse M2 bolts at Y=-320, -302, -282 and -264, Z=-37.964. Use washers and locking nuts. Inner bolts pass through A37; outer bolts clamp the wood stack.
3. Fasten A09 to A37. Fit the **4 mm spring-wire shaft** through the finished 4.2 mm bores. Fit the 4 mm clamp steering arm and retainers. Keep the shaft turning freely.
4. The wire reaches Z=+1. Trim only after checking full clamp engagement. Use mechanical retention, not glue. Fit the 76.2 mm nose wheel without squeezing its hub.
5. Keep the steering horn below the deck and check its full travel. Make the 6 mm skin clearance hole only after forming the lower nose skin. Recheck under load and steering deflection.

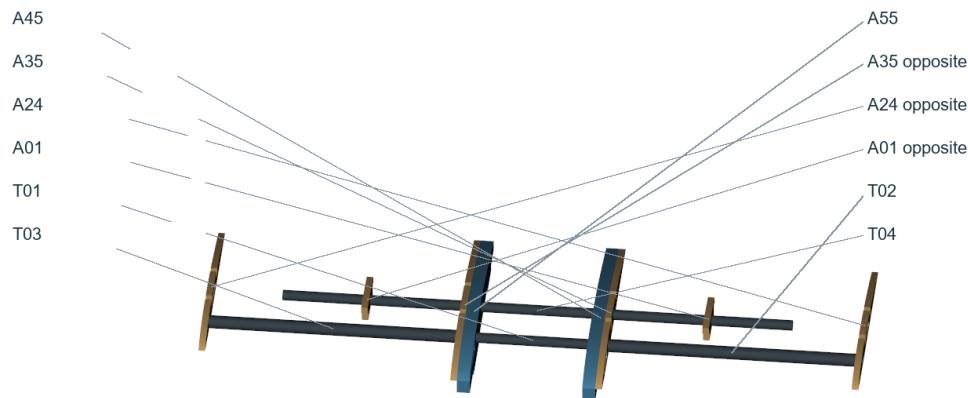
Colours identify parts; they do not specify paint or material. The cover, GPS and pitot attach independently of the steering shaft. Use nonmagnetic fasteners near the compass.

05 / Wing frames and controls



1. Build the right wing and its left-hand mirror. Fit A35, A24 and A44 to A05/A07. Fit A01 and the front sleeve. Use the part quantities, not the number of source-side examples.
2. Jig both wings straight and check equal incidence. Finish designated rib/end-plate edges to their assembly contour while preserving bores and slots.
3. Fit A53/A10, hinge rail A36 and aileron A21 on the right; A23/A11 on the left. Fit smooth 3 mm pins, retain both ends and leave about 1 mm axial freedom.
4. Mount an MG90S in each A44 opening using two M2 bolts, broad washers and locking nuts. The body opening is 23.4 x13.0 mm; verify the actual servo before cutting both ribs.
5. Set servos and controls neutral. Put each horn pin over its hinge axis and align its plane with the servo arm. Use H05B backing plates and through-fasteners. Add crush-resistant sleeves through foam or hollow printed controls.
6. Cut pushrods after fitting both rod ends. Fit retention clips. Check full combined travel and servo-arm sweep with the skin temporarily fitted; make a local service/rod exit opening only where needed.

06 / Joiners and wing retention



Tube	Section and length	Assembly position
T01	8 OD /6 ID; 406 mm	X=-203..203; axis Y60.454, Z15.632
T02/T03	10 OD /8.2 nominal ID; 199.35 mm each	Right X53..252.35; left mirror
T04	8 OD /6 ID; 390 mm	X=-195..195; axis Y100.500, Z15.070

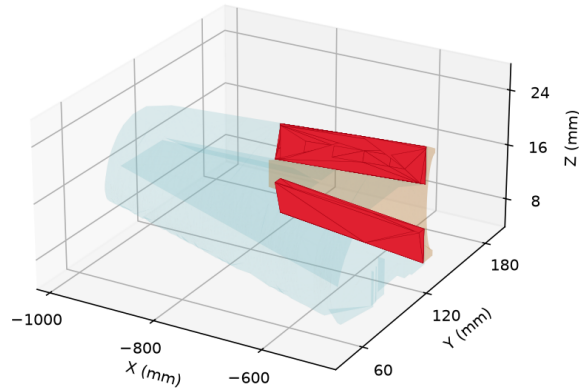
Use a measured matched sliding pair. The 5 mm solid carbon rods are spare material, not the specified joiners. Do not force a tube or sand away its structural wall.

Centre both joiners and test both wings together. Bond sleeves only, keeping adhesive off the sliding joiner. Mark the insertion positions; front engagement is about 150 mm per wing.

Fit three M2 root retainers per wing with washers and locking nuts. Check twist, spanwise escape, wiring disconnects and hinge clearance. Label the left/right servo connectors and strain-relieve them.

07 / Forming the foam

Wing hinge inner core | red = CAD interference to relieve

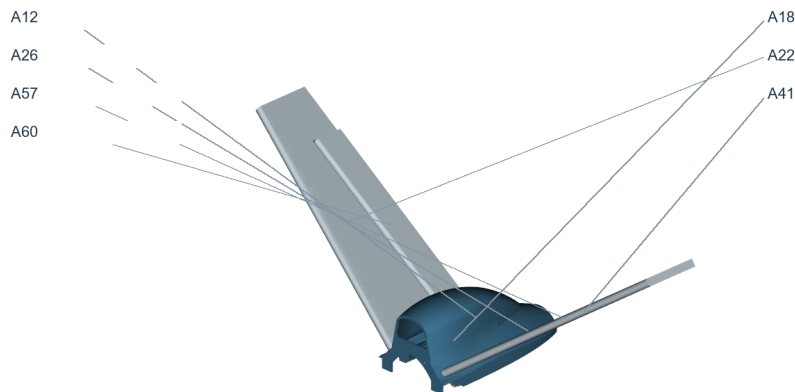


Blue: A27 foam. Tan: A23. Red: trim contact region. View cropped; axes not equal scale.

1. Print at Actual Size /100%. Measure the 100 x20 mm rectangle. Join only matching tile marks with the specified overlap.
2. Mark OUTSIDE and the F-series ID. Keep the outside paper as the developed pattern reference. Test the tightest curve on scrap first.
3. Peel inside paper only in curved zones. Relieve/compress the core gradually over the actual frame. Do not stretch a short edge to close a seam.
4. Bevel inside edges to the actual meeting angle. Butt outside-paper edges; use separate 15 mm internal paper straps. Keep glue and straps clear of moving parts.
5. Dry-fit a whole wing and its control before bonding. Hold alignment through cure. Use adhesive approved for XPS; ordinary CA must not reach exposed foam.

CUT lines are finished skin edges. F23/F24 alone have a 1 mm paper-only lower lip. F31/F32 each need two layers and have 5 mm sacrificial shaping stock. Other panels have no added lap or trim.

08 / Tail and service openings



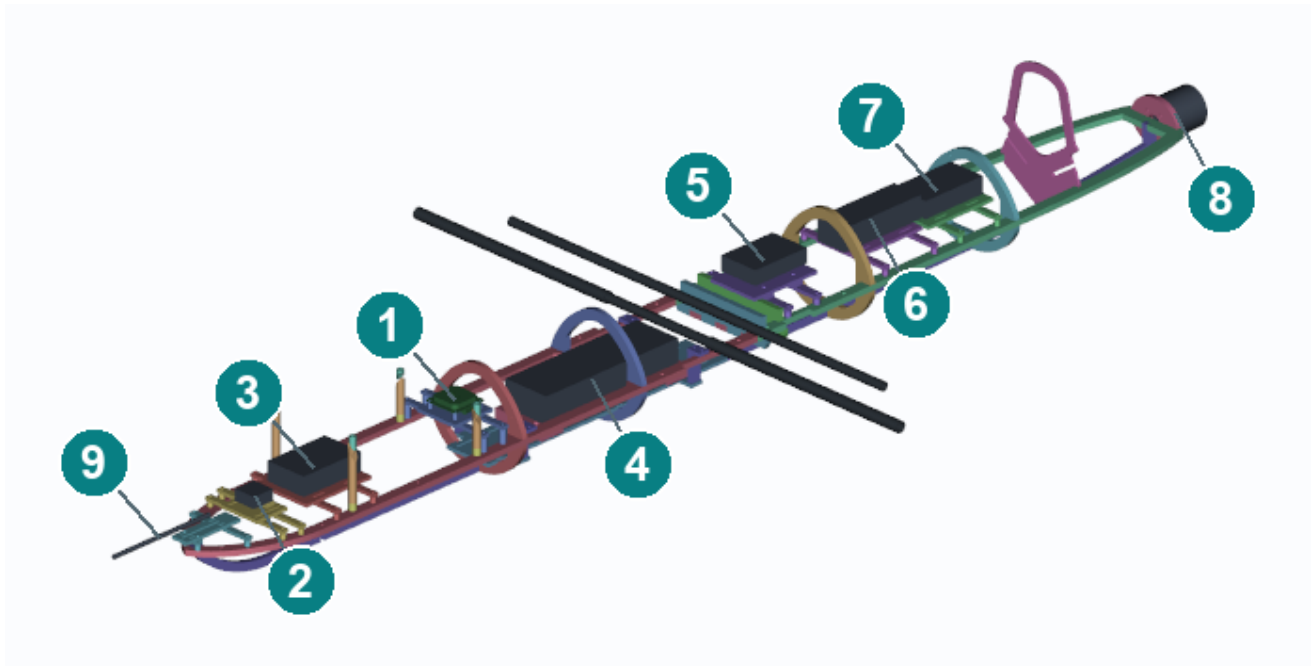
Form the fixed tails with the split collars. Check their fit in A12 and equal angles before bonding. Laminate each moving tail from two 5 mm layers, trim to the finished outline and shape long edges to R5.

Fit both MG90S servos in A12 using the slotted ear holes. Tape-hinge the controls on the indicated line. Fit backed horns and adjustable pushrods; check combined pitch/yaw travel without binding.

Location	Finish after forming
F25/F26 forward roof	Join the two skins, then make the 50 x100 mm plan-view service opening: X +/-25, Y=-115..-15. Retain the lid with paper edge supports and replaceable reinforced tape. Keep A54 intact.
F23/F24 rear roof	Join as one removable cover. Add 15 mm internal paper supports and six reinforced-tape retention tabs. Do not glue this cover permanently shut.
Cooling	Inlet: 24 x12 mm at Y250. Outlet: 30 x20 mm at Y374. Add a small forward-facing scrap-foam scoop over the inlet to direct air toward the ESC. Verify cooling by test.
Gear exits	Form skins first. Mark actual gear crossings; provide 3 mm nominal clearance around the main legs and a 6 mm nose-shaft hole. Reinforce the paper edges and verify loaded clearance.

These openings are finishing operations on the formed skins. Do not use their plan-view dimensions as replacement flat foam patterns.

09 / Equipment mounting map

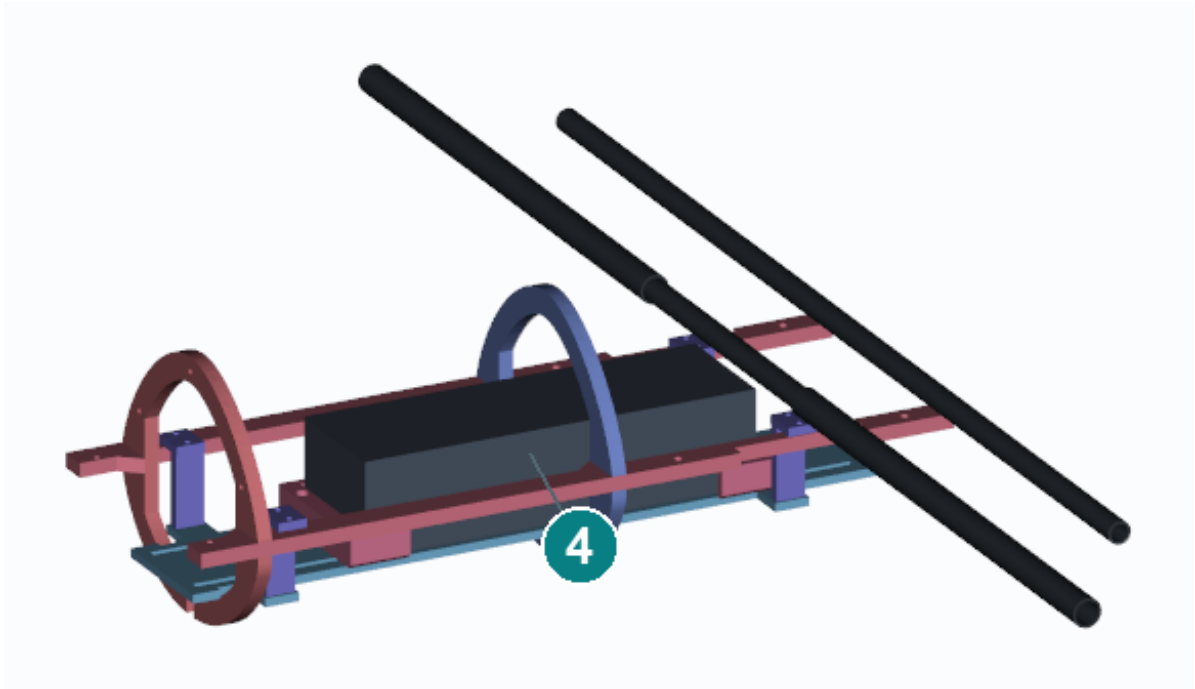


Mark	Equipment	Station from nose / mounting
1	M9N-5883	286.65 mm; I09 inside nose on A13
2	ASPD-4525	76.65 mm; I13 on A13
3	FS-iA10B	139.65 mm; I10 on A13
4	GenX battery	Centre adjustable 379.65..459.65 mm; I01 below deck
5	F405-WING V2	609.65 mm; I04, 30.5 mm grid
6	40 A ESC	719.65 mm; I12 on A43
7	SiK air radio	791.65 mm; I11 on A43
8	Motor	A26 mounting face at 1022.30 mm
9	Pitot	I14; pressure ports start at least 50 mm ahead of the nose

Install the battery with main and balance leads exiting toward the tail. Keep nearby power wiring aft of Y=-37, at least 100 mm behind the GPS module. Never route a battery lead across the internal GPS bridge.

Equipment shapes in illustrations are placement references. Use the actual connectors and secure every item with the specified fasteners/straps. Route wires along the frame sides, away from the battery insertion path.

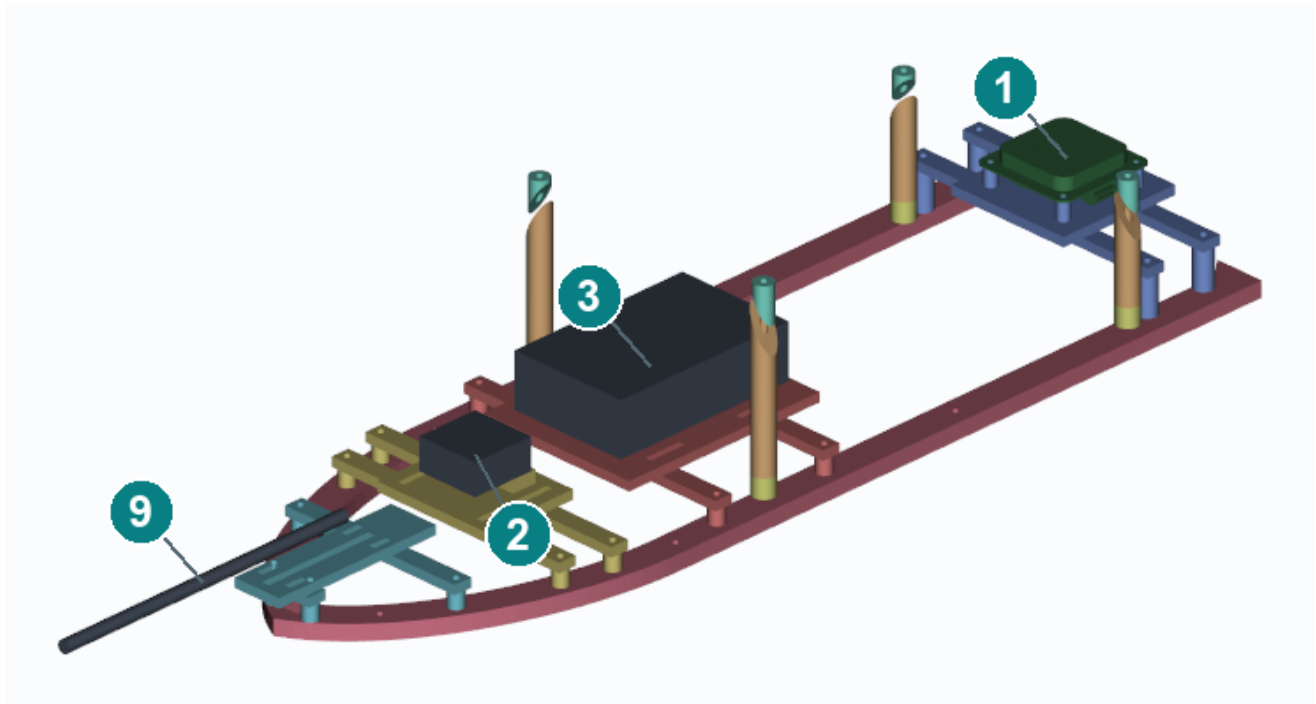
10 / Battery tray and access



1. Bolt four I17 hangers to A13 at X +/-40, Y=-115 and 65. Bolt I01 below them at X +/-32. Tray underside is Z=-28. Its local edge notches clear A34/A54; do not cut those formers.
2. Fit a 1 mm non-slip pad. The 138 x43 x33 mm pack sits with its bottom at Z=-24. Confirm clearance to both tubes with the actual pack; no wiring may pass over the pack in this gap.
3. Fit I02 corner stops with their fingers toward the pack. Their centres are 73 mm ahead of and behind the selected battery centre. Use M2 screws, washers and locking nuts in the slots.
4. Remove the front stop to load. With the nose cover removed, place the pack forward of A34 and slide it aft into the tray. Fit the front stop and two separate 20 mm straps around both pack and tray. Keep stops padded; never squeeze the pouch.
5. Set the initial location during dry assembly, then balance the finished aircraft. Available centre travel is Y=-60..+20. Use a ball-end hex driver and long-nose pliers for fasteners under a joiner; verify access before closing skins.

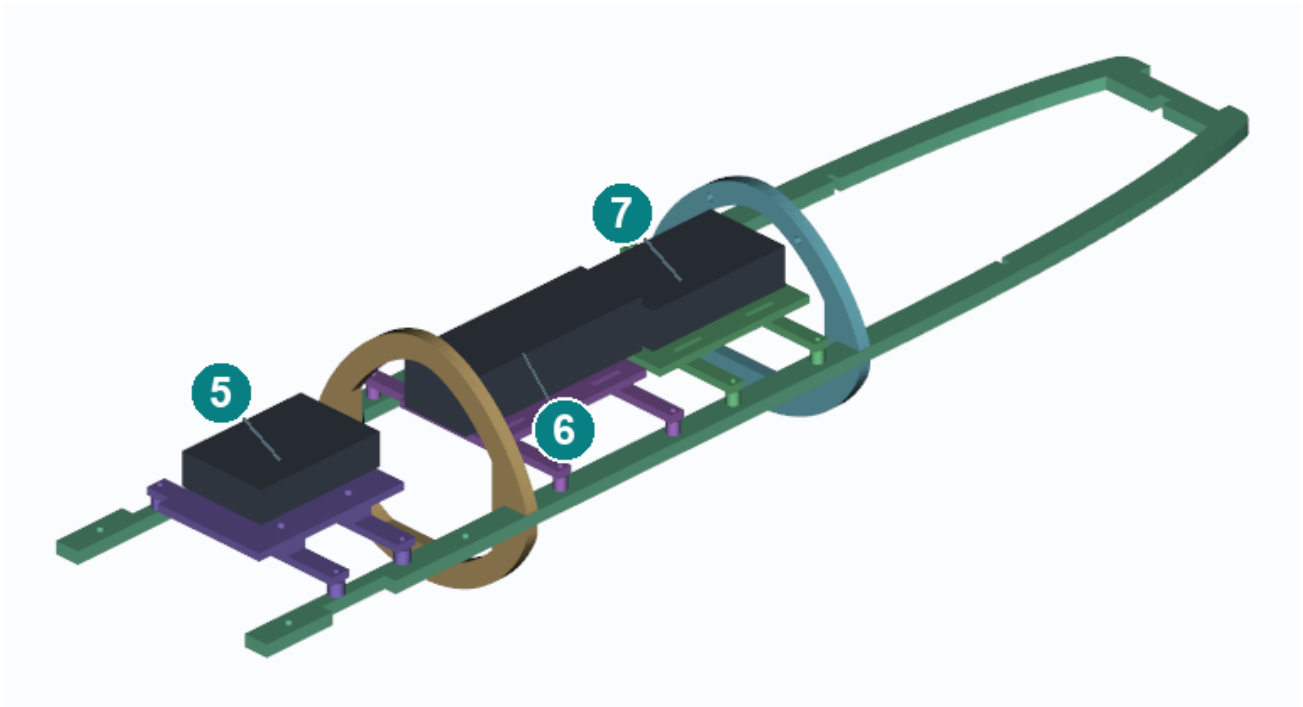
The F25/F26 opening and front A06 hatch give access to straps and stops. Support tabs carry the printed hatch; two replaceable 20 mm reinforced-tape strips retain it.

11 / Nose sensors and cover



1. Bolt I14 to A13. Fasten the supplied pitot tab in the centre slot with an M2 bolt, washers and locking nut; add the anti-rotation strap. Align the tube forward, parallel to the fuselage centreline.
2. Start with every pressure port at least 50 mm ahead of the nose. Keep ports clear of straps and adhesive. Connect both tubes to I13 without kinks; identify total and static pressure ports from the sensor/kit markings.
3. Fit ASPD-4525 with two M2 screws on its 15 mm hole spacing. Insulate the board. Keep the pressure tubing and I2C cable accessible under the nose cover.
4. Bolt I09 to A13 at Y=-153 inside the nose, using four M2 nylon feet. Mount M9N-5883 level, antenna up and arrow forward on its 26 mm grid. Keep bolt tips above Z11. Route battery power leads aft; use nylon fittings nearby and check compass interference with the motor running.
5. Fit I10 and strap the receiver to its insulating pad. Support the two active antenna tips independently at right angles, clear of carbon and power wiring. Check range with the installation complete.
6. I00 six-mm spacers support the four I05 columns; Compact I06 washers match their curved cover seats. Use nylon cover bolts near the compass. Front columns are at Y=-300, rear at Y=-170. Remove four cover bolts and lift the cover. GPS, receiver and pitot remain on the frame; keep their wires and tubes clear.

12 / FC, ESC and telemetry



1. Bolt I04 to A43. Mount the F405-WING V2 on its 30.5 mm grid with 6 mm nylon spacers and supplied grommets. Do not crush the grommets. Keep USB, SD and connectors accessible through the rear A06 hatch.

2. Strap the ESC to I12 and the air radio to I11 over insulating pads. Leave connectors and the ESC cooling face clear. Straps must restrain the cases without loading a PCB connector.

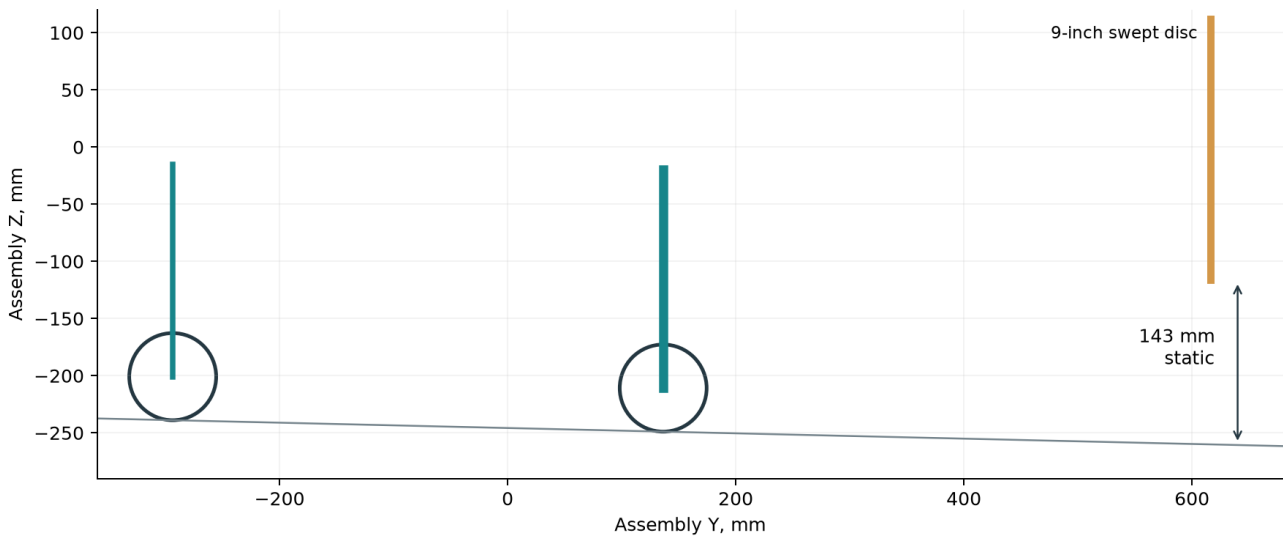
3. Keep the FC-to-ESC battery-power run short. Route motor phase extensions aft along the frame side, away from radio/sensor cables. Use appropriate flexible copper wire and support every connector; do not extend battery leads casually.

4. Support the 433 MHz antenna separately from its connector. Fit the correct antennas to both radio units before power. The purchase listing conflicts on power; use the actual radio label and firmware limits, with 100 mW as the requested setup.

5. Fit the removable F23/F24 cover and cooling inlet/outlet. Verify wire clearance, connector access and measured ESC temperature under load.

The Electronics Team guide gives pad-to-pad wiring, BEC isolation, output assignments and calibration. The four rear-bay carrier foot holes are already in A43 for each component.

13 / Landing gear for grass



G01: 20 x4 mm 6061-T6 strip, 12 mm inside bend radius. Use the formed STEP as the dimensional authority. The K=0.40 flat DXF is a tooling starting point: qualify a bend coupon and adjust allowance before cutting the final blank. Reject cracks; do not soften the part without restoring and verifying the required material condition.

Bolt G01 below G03/A47 with two M3 through-bolts and broad washers. Use two smooth-shank M4 wheel axles. Fit three lightweight 76.2 mm wheels with 4 mm bores; adapt spacers to measured hubs.

G02 uses 4 mm spring wire formed by a suitable shop. Nose wheel reference: 22 mm hub width, centre X26. Allow for the inner collar/washer, wheel, outer washer/collar and positive retention on the 46 mm axle.

9-inch propeller clearance	Calculated rigid geometry
Static, flat ground	143 mm
10 degrees nose-up from static	59 mm
Same 10-degree case, minus assumed 15 mm deflection and 25 mm grass	19 mm remaining

These are design cases, not measured gear deflection or a landing rating. Check the assembled gear at measured aircraft mass, including fore-aft loads, joint motion, wheel freedom and skin clearance. Use short, even grass; the figures do not cover rough ground or arbitrary flare angles.

14 / Prints, motor and finishing

Group	Starting process; qualify actual material
A06/A59 shells and access lids	Available PLA+ is not LW-PLA. Start 0.2 mm layers, 2 perimeters and 3-5% infill; inspect thin walls and record slicer/finished mass.
A12 structural tail housing	Carries tail-root and servo loads. Qualify local slot walls, layer orientation, root bonds and the A12-to-A18 connection. The 2-wall/5% slicing case is a mass estimate, not an accepted production recipe.
Ailerons and hinge rails	Start 3 perimeters, 10% infill. Keep supports out of bores; fit smooth pins and verify full motion.
Load-bearing mounts and carriers	Start 4 perimeters, 30% infill, with solid material around bolts. Orient loads to avoid splitting layer interfaces. Test the actual material/process.

Print 42 installed pieces from 38 STL files, using the quantity table. The fit coupon is extra. Files are millimetres at 100%; some long parts need a larger printer. Do not make an arbitrary split or rescale.

Clear all passages, measure the coupon and inspect layer bonds. Record every finished print mass. Slicer settings alone do not establish strength or heat resistance.

A26 motor mount: four 3.2 mm holes on a 19 mm pitch circle; adjacent centres 13.435 mm. Preserve the 12 mm centre opening. A62R ends flush with the rear mounting face at Y582.65; no spine tab extends into the motor. Measure screw engagement so no screw can reach windings.

Use the selected Emax 2807 / 9-inch three-blade / 4S /40 A combination as the design assumption. Current, thrust and temperature are still measured before flight. Keep the propeller removed during assembly and configuration.

Body assembly still needs foam forming, adhesive, fasteners and linkage adjustment after the machine work. Weigh the finished body before the Electronics Team chooses the battery position.

15 / Body acceptance and transfer

Check	Record on the actual aircraft
Stock and scale	Measured ply/foam thickness; slot coupon; 100 x20 mm rectangle; printer dimensions.
Structure	All 37 wood pieces present; correct grain; bonded joints and bypass fasteners; straight frame and matched wings.
Controls	Five MG90S secure; retained pins/rod ends; backed horns; full combined motion without binding or servo stall.
Battery and access	Actual pack fits and can be removed; both straps and stops restrain it; tubes and cables remain clear.
Gear	Material/process accepted by shop; wheels and steering free; load/deflection and attachment checks recorded.
Skins	No forced stretching, open seams or twist; reinforced gear exits and removable covers secure.
Electronics interfaces	Pull strings, labelled wing connectors, pitot tubes, independent antenna support and service access.
Mass	Body and individual print masses recorded; include glue, fasteners, wiring and finish.

Balance reference: 522.3 mm aft of the nose, or 22.2 mm aft of the front joiner, is the geometric quarter-MAC. It is not an approved flight CG range. Use the measured mass worksheet and final complete-aircraft balance.

Battery movement changes CG by battery mass / total mass times its travel. The available 80 mm travel is useful trim; do not assume it can correct any weight distribution.

Before hand launch, the Electronics Team and an experienced fixed-wing pilot must review the finished mass, CG, control authority, propulsion and launch procedure. Complete the prop-removed launch-detection, range and failsafe tests first.